

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Noise pollution - Set 1

Mark scheme

AQA spec 3.4.3.2.13; Pollution; textbook pp. 286-295

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) • A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity	2
01.2	• Difference = 30 dB = 3 x 10 dB • = 10^3 = 1000 times	2
01.3	• 88 dB • Doubling the sound intensity increases the level by 3 dB (logarithmic scale)	2
01.4	• 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing; quieter sounds exist but cannot be heard	1
01.5	• 20 to 20 000 Hz • dB(A) weights sounds between about 1000 and 4000 Hz more heavily • Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance; range narrows with age or noise damage	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	
02.1	<i>Any 3 from:</i> • Stress-related illness: ulcers, high blood pressure, heart disease • Behavioural changes: irritability, aggression, lack of concentration leading to accidents • Communication problems / educational underperformance (eg schools under flight paths) • Sleep disturbance	3

Question	Marking guidance	Marks
02.2	• Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) • Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes • Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry	3
02.3	• Birds that sing to establish territories / attract mates cannot be heard against background noise • Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea	2
02.4	• Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source • Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	
03.1	• A turbofan at the front forces bypass air around the inner cowlings / core jet • The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise • The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest)	3
03.2	<i>Any 2 from:</i> • Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly • Acoustic liners in the engine cowlings absorb noise • Hush kits fitted to older low-bypass engines • Fairings on undercarriage and wing-flap hinges reduce turbulence; welded rather than riveted panels • Lighter composite aircraft need less thrust • Blended wing with engines on top so the body shields the ground	2
03.3	• Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer • Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low • Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft • More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters	4
03.4	• Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	
04.1	• Wheel vibration: track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension • Pantograph: aerodynamic fairing • Wheel squeal: lubrication of wheels / track on corners • Braking: composite-material brakes instead of cast iron	4

Question	Marking guidance	Marks
04.2	Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas	1
04.3	<i>Any 3 from:</i> - Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) - Acoustic fences, embankments / baffle mounds, walls or dense tree planting - Traffic management for free flow (less acceleration noise); re-routing traffic away from residential areas - Double glazing / acoustic insulation of buildings - Vehicle design: better aerodynamics, engine insulation, quieter exhausts, quieter tyres	3
04.4	<i>Any 2 from:</i> - Air bubble curtains around the pile driver absorb noise - Acoustic survey to detect cetaceans first and delay work if present; increase volume gradually (soft start) so animals can leave - Drilling instead of pile-driving	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	
05.1	- Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier - Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground - Take repeated readings at each distance and calculate means (and standard deviation) to assess variability - Use a measure such as LA10 or Leq over a set period rather than a single peak reading - Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance	5
05.2	Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier	1
05.3	- LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods - LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level	2
05.4	<i>Any 2 from:</i> - Double / triple glazing - Acoustic insulation in walls / roof - Bedrooms on the side facing away from the track - Dense hedges / trees and earth mounds in gardens	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
06.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
06.2	• The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	